

# GCSE algebra - Set 2 - Solutions

Specification M4.1-M4.8, M4.18, M4.19.

## Answer key

| Question | Answer | Topic        |
|----------|--------|--------------|
| 1        | C      | GCSE algebra |
| 2        | A      | GCSE algebra |
| 3        | E      | GCSE algebra |

## Worked solutions

### Question 1 ALG-15 · GCSE algebra · spec M4.18

The distance-time graph of a journey is a straight line from (0, 0) to (2, 60), then horizontal to (3, 60), then a straight line to (5, 0) (time in hours, distance from home in km). What was the average speed for the whole journey?

**Answer: C**

Total distance  $60 + 60 = 120$  km in 5 hours: 24 km/h. (Displacement is 0 but distance travelled is not.)

### Question 2 ALG-03 · GCSE algebra · spec M4.6

Simplify  $\frac{x^2 - 9}{x^2 + x - 12} \times \frac{x^2 - 16}{x^2 - 3x}$ .

**Answer: A**

Factorise everything:  $\frac{(x-3)(x+3)}{(x+3)(x-4)} \times \frac{(x-4)(x+4)}{(x+4)(x-3)}$ . The  $(x+4)$  factors cancel and one  $(x-3)$  cancels, leaving  $\frac{(x+3)}{x(x-3)}$ .

### Question 3 ALG-08 · GCSE algebra · spec M4.19

The  $n$ th term of a sequence is  $\frac{n^2 + 1}{2n}$ . Which of the following statements is/are true?

- I The 5th term is 2.6.
- II Every term after the first is greater than the previous term.
- III Some term of the sequence equals 10.

**Answer: E**

I:  $\frac{26}{10} = 2.6$ . II:  $u_n = \frac{n}{2} + \frac{1}{2n}$ ;  $u_1 = 1$ ,  $u_2 = 1.25$ , and for  $n \geq 1$  the increase  $\frac{1}{2} - \frac{1}{2n(n+1)} > 0$ , so increasing. III:  $n^2 - 20n + 1 = 0$  has no integer solutions (discriminant 396 is not a square), so false.